

SUMMARY

Results-driven **Software Developer** with a **Bachelor of Science in Computer Engineering** and a proven track record of taking digital health systems from design to production at the Infectious Diseases Institute. Full-stack proficiency in **FastAPI, React, JavaScript, Python (Django), Kotlin (Android), Java/SpringBoot, OpenMRS, Docker, and Linux**, paired with hands-on experience in **IVR and PBX** systems, and CI/CD-driven deployment automation. Has independently led production releases including a multi-language CFL call flow and a 15+ report vxnaid suite built a team-wide documentation platform. Combines strong engineering fundamentals with a builder’s instinct for clean, maintainable, user-centric systems, and a continued interest in cybersecurity and emerging AI tooling.

TECHNICAL SKILLS

Programming Languages: Java, Kotlin, Python, JavaScript, TypeScript, SQL
Frontend: Reactjs, Nextjs, Bootstrap, Tailwind CSS
Backend: Django, SpringBoot, Expressjs, FastAPI
Mobile App Development: Kotlin, Java, Flutter
Databases: MySQL, PostgreSQL, MongoDB
Health Systems: OpenMRS, VxNAID, Call for Life (CFL)
Telephony & IVR: Asterisk, PBX Systems, SIP, AGI, CallFlow/IVR Design
DevOps & Infrastructure: Docker, GitHub Actions (CI/CD), Linux, On-prem Deployments, AWS
Tools: Git, GitHub, Docker, Linux, Figma, Microsoft Office Suite, Adobe (*Illustrator, Photoshop*)

EMPLOYMENT HISTORY

Junior Software Developer, Infectious Diseases Institute (IDI)	Oct 2025 — Present
✓ Led production deployment of the vxnaid Visit Reminder call flow with Luganda-language support. ✓ Delivered 15+ vxnaid web and app reports while fixing production bugs in sync, validation. ✓ Developed IVR/AGI call flows and PBX system setup for the HASH conversational AI agent. ✓ Migrated the staging environment to on-premise and built a production replica environment. ✓ Configured and maintained Linux servers and on-prem deployments for vxnaid and CFL. ✓ Automated deployments and built CI/CD workflows using Docker and GitHub Actions. ✓ Designed and built a GitHub Pages documentation platform centralizing technical.	
Graduate Software Developer, Infectious Diseases Institute (IDI)	Jan 2025 — Sept 2025
✓ Supported development and testing of the VxNAID Android and web applications. ✓ Built and fixed OpenMRS reports, messaging workflows, and data validation issues. ✓ Installed, configured, and tested a local Asterisk server for SIP-based IVR call handling. ✓ Set up tablets, facility credentials, and local development environments. ✓ Debugged offline sync, visit scheduling, and reporting inconsistencies. ✓ Assisted with deployments, user training, and field support across health facilities.	
Software Developer Intern, NetLabs!UG	June 2023 — Sept 2023
✓ Learned core machine learning concepts, including data preprocessing and model evaluation. ✓ Designed a smart irrigation system using Python and TensorFlow to optimize water usage. ✓ Integrated sensor data into an embedded system for real-time decision-making.	

- ✓ Acquired practical experience in embedded systems, networking, Python, and machine learning.

Junior Software Developer Intern, Zidio

August 2024 — Oct 2024

- ✓ Analyzed project data and produced summaries that supported team decision-making.
- ✓ Cleaned and transformed raw datasets to ensure data quality for downstream analysis.
- ✓ Built backend services in JavaScript, improving server performance and scalability.
- ✓ Designed REST APIs and implemented secure database integrations.

Software Developer Intern, EvoTech Solutions

June 2022 — Sept 2022

- ✓ Built UI components for a school management system using React.
- ✓ Implemented interactive features that improved usability and performance.
- ✓ Collaborated with the team to optimize frontend code and rendering performance.
- ✓ Authored and managed system content to improve accessibility and clarity for end users.

EDUCATION

Bachelor of Science in Computer Engineering

Makerere University, Kampala, Uganda

Uganda Advanced Certificate of Education

St. Peter's College, Tororo

Uganda Certificate of Education

St. Peter's College, Tororo

SOFT SKILLS

Problem-Solving Adaptability Communication Teamwork Time Management Quick Learner

PROJECTS

Asterisk Audio Generator

Python, FastAPI, Kokoro TTS, sox, Sunbird

Built a web tool that generates Asterisk-compatible 8kHz mono MP3 prompts from text across three TTS backends (Kokoro, Sunbird AI, edge-tts), supporting English, Luganda, and Swahili. Designed single and batch CSV generation workflows with live progress via Server-Sent Events, a REST API, and a one-shot installer for local model setup.

VxNAID — Vaccine & Immunization Management System

Kotlin, OpenMRS

Developed and supported immunization workflows, reporting modules, and IVR integrations. Fixed production bugs, improved data accuracy, deployed updates to staging and production, and trained health workers across multiple facilities.

Voice-Controlled Wheelchair

Python, Embedded Systems, TensorFlow

Engineered a voice-activated wheelchair system using speech recognition, enabling hands-free control for users with physical impairments significantly improving accessibility.

REFERENCES

Names, Contacts and Emails available on request